



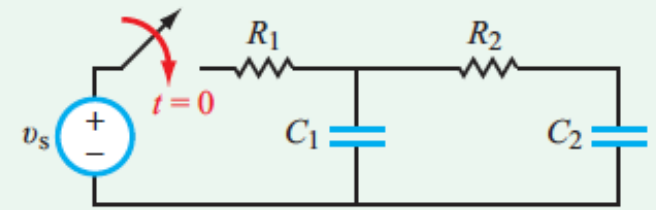
Lecture 7

- Second-Order Circuits

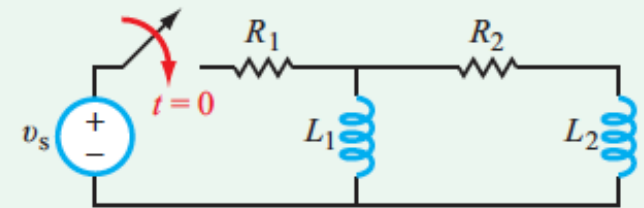
Second-Order Circuits

- Two energy storage elements
- Analysis: Determine voltage or current as a function of time
- Initial/final values of voltage/current, *and their derivatives* are needed

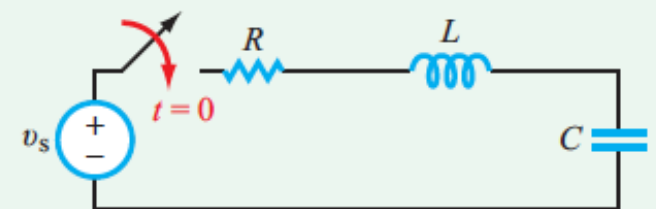
A second order circuit is characterized by a second order differential equation.



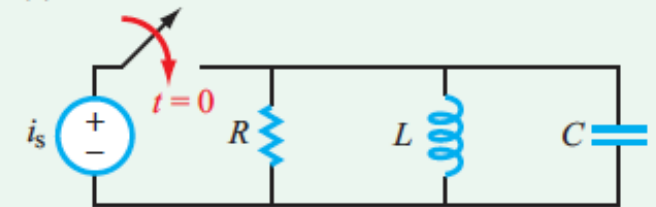
(a) 2 capacitors



(b) 2 inductors



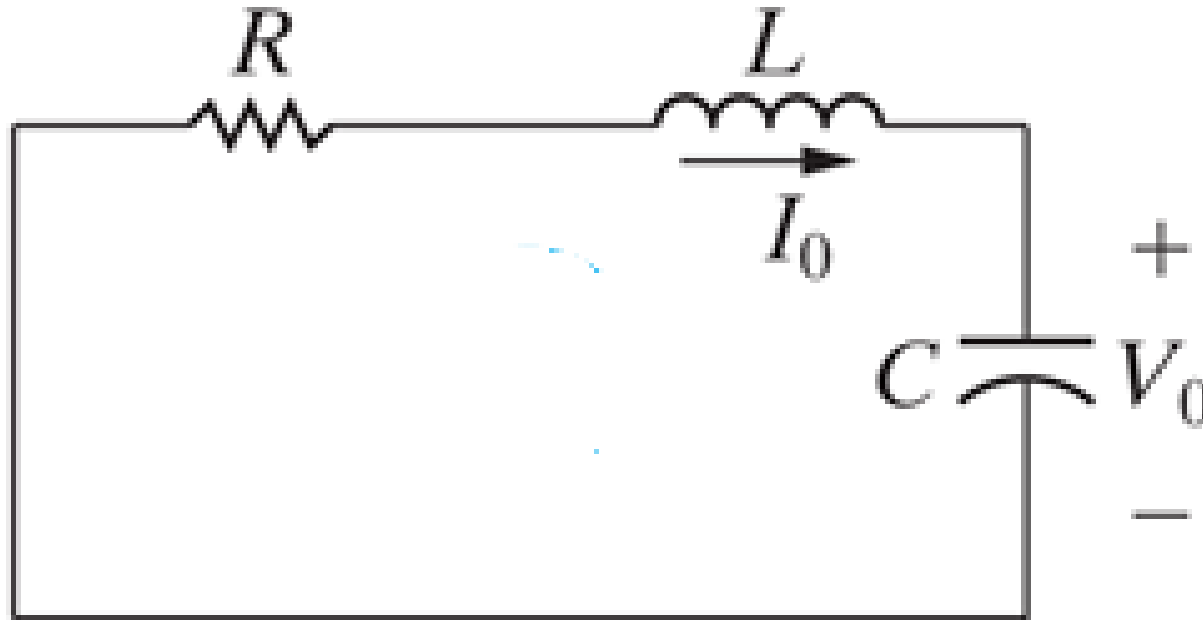
(c) Series RLC



(d) Parallel RLC



Source-Free Series RLC

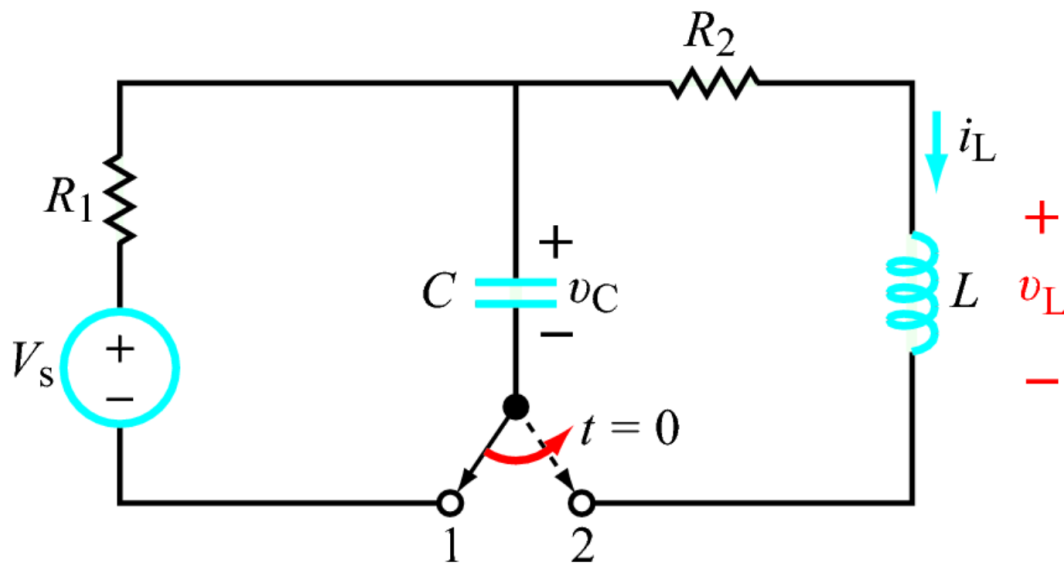


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$$\frac{d^2 v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{1}{LC} v = 0$$



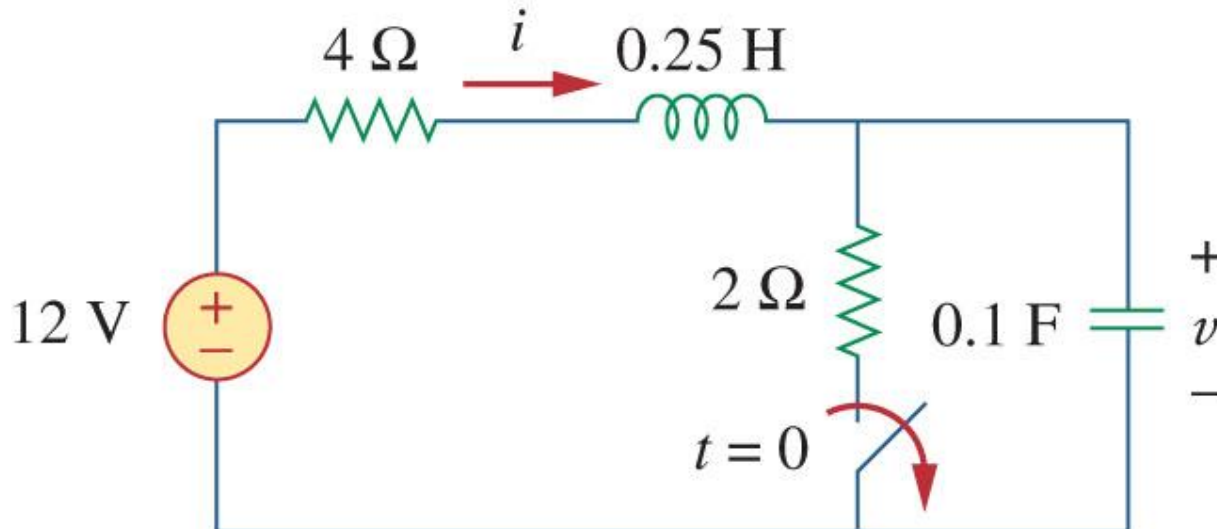
Initial and Final Conditions



Example

- The switch has been closed for a long time. It is open at $t = 0$. Find
 - $i(0^+)$, $v(0^+)$
 - $di(0^+)/dt$, $dv(0^+)/dt$
 - $i(\infty)$, $v(\infty)$

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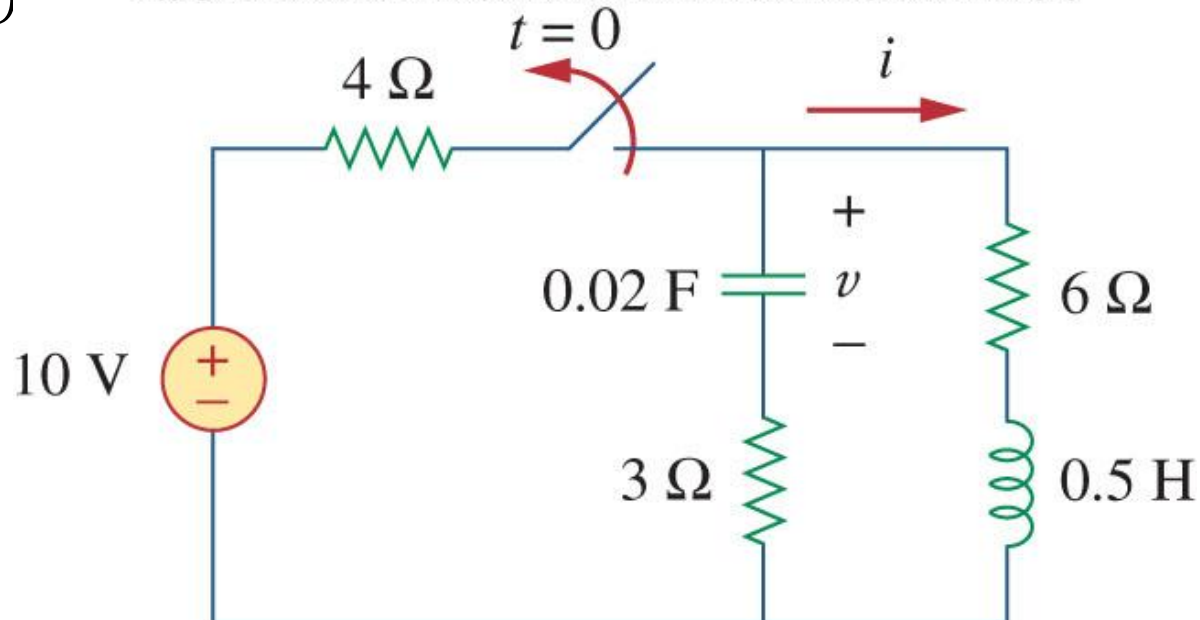
Exercise

- Assume the circuit has reached steady state at $t = 0^-$.

Find

- $i(0^+), v(0^+)$
- $di(0^+)/dt, dv(0^+)/dt$
- $i(\infty), v(\infty)$

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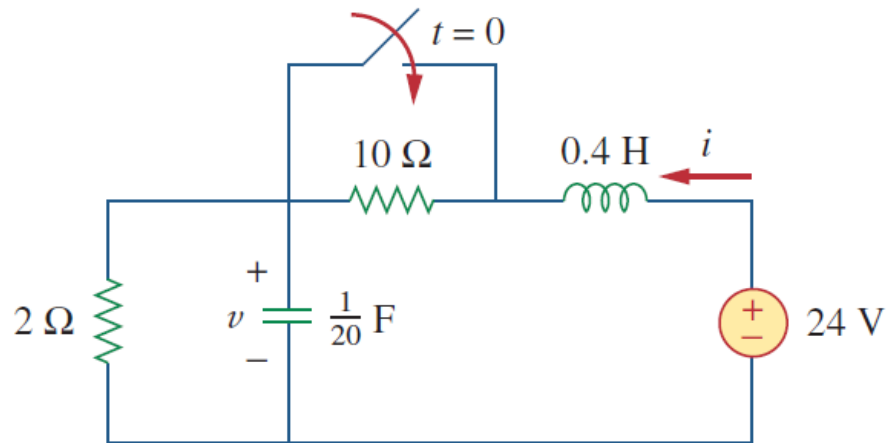


Figure 8.4

For Practice Prob. 8.1.



For the circuit in Fig. 8.7, find: (a) $i_L(0^+)$, $v_C(0^+)$, $v_R(0^+)$,
(b) $di_L(0^+)/dt$, $dv_C(0^+)/dt$, $dv_R(0^+)/dt$, (c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$.

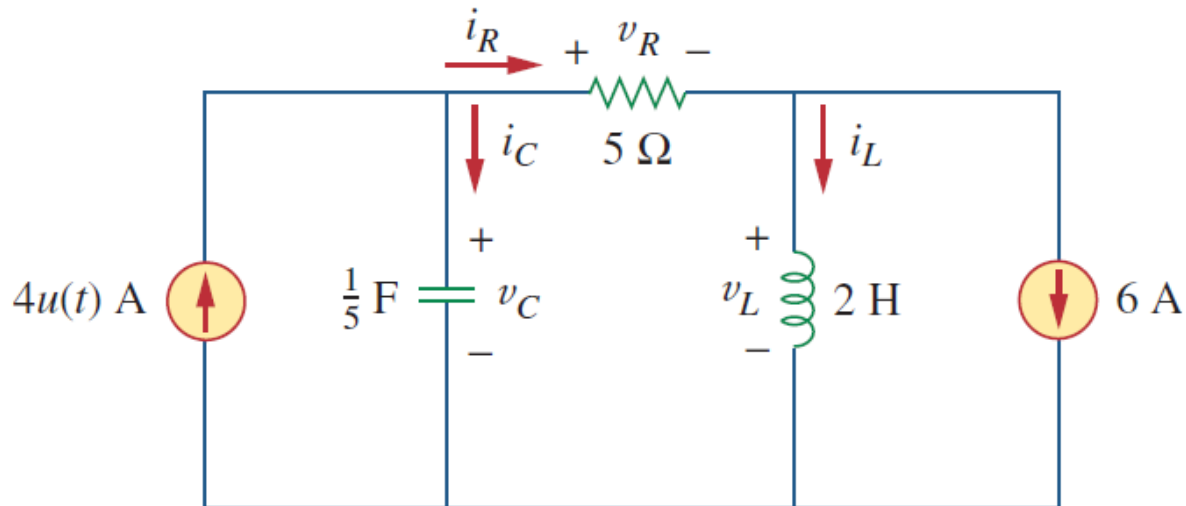
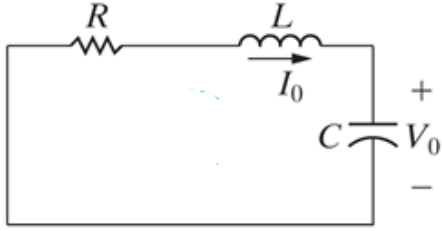


Figure 8.7

For Practice Prob. 8.2.



Source-Free Series RLC



$$\frac{d^2 v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

$$\alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

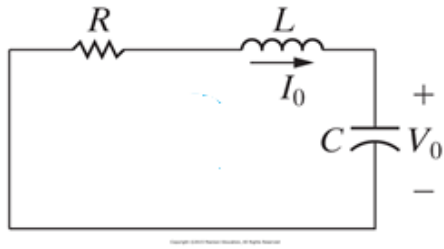


$$s = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$\alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_o^2} \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_o^2}$$

Case 1: Overdamped ($\alpha > \omega_0$)

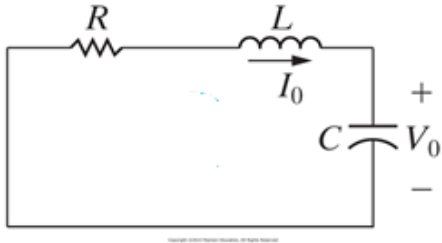


$$\alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Case 2: Critically Damped ($\alpha = \omega_0$)



$$\frac{d^2 v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{1}{LC} v = 0 \quad s = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$\alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$s_1 = s_2 = -\alpha = -\frac{R}{2L}$$

Case 2: Critically Damped ($\alpha = \omega_0$)

Go back to

$$\frac{d^2 v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{1}{LC} v = 0 \quad \alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

- When $\alpha = \omega_0 = R/2L$, the equation becomes

$$\frac{d^2 v}{dt^2} + 2\alpha \frac{dv}{dt} + \alpha^2 v = 0$$

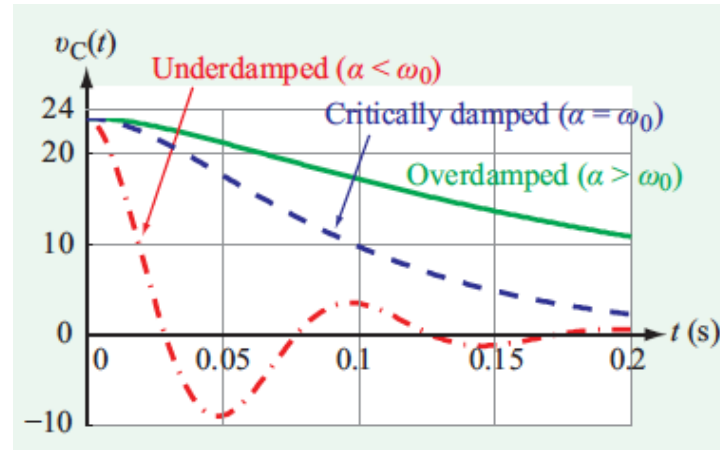


$$\frac{d}{dt} \left(\frac{dv}{dt} + \alpha v \right) + \alpha \left(\frac{dv}{dt} + \alpha v \right) = 0$$

- $v(t) = (A_1 t + A_2) e^{-\alpha t}$

Case 2: Critically Damped ($\alpha = \omega_0$)

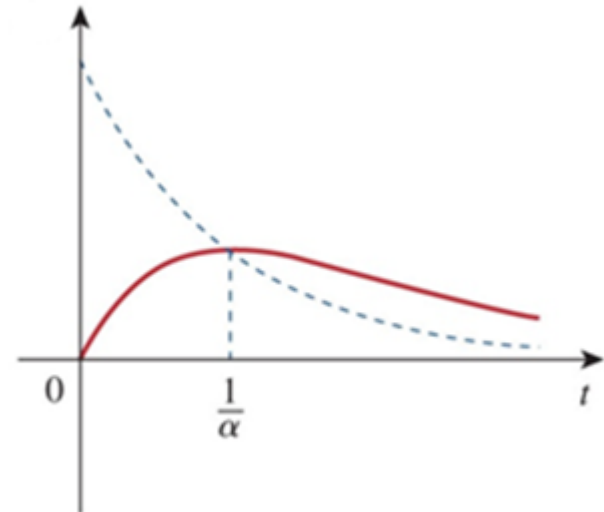
$$v(t) = (A_1 t + A_2)e^{-\alpha t}$$



(If $A_2 = 0, A_1 = 1$)

A typical critically damped response is shown

Why maximize at $t = \frac{1}{\alpha}$?



Case 3: Underdamped ($\alpha < \omega_0$)

$$s = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -\alpha + \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha + j\omega_d$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -\alpha - \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha - j\omega_d$$

where $j = \sqrt{-1}$ and $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$.

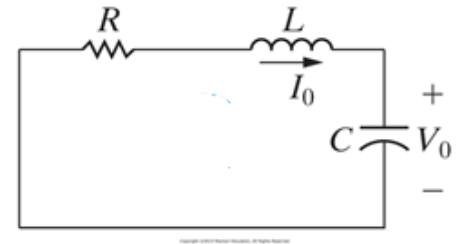
- ω_0 is often called the undamped natural frequency.
- ω_d is called the damped natural frequency.

The natural response

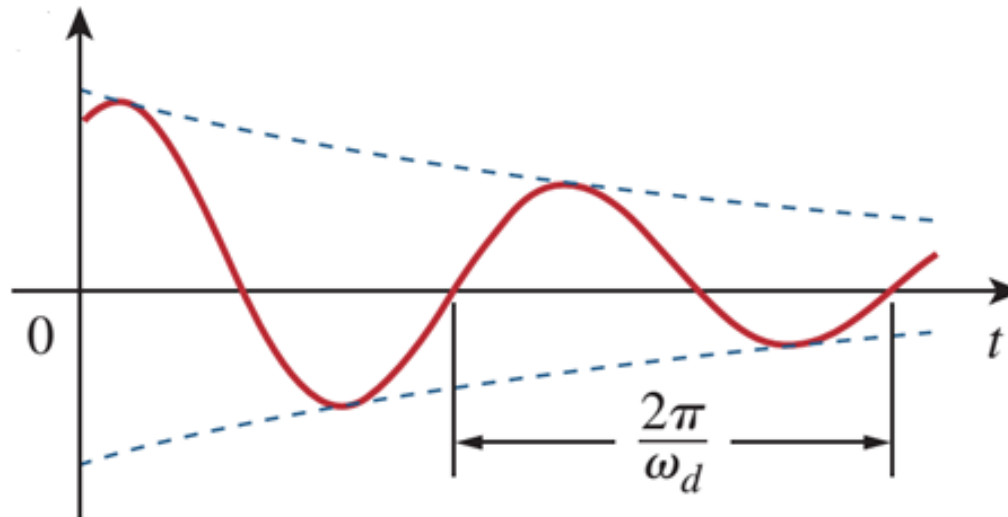
$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$



- Exponential $e^{-\alpha t}$ * Sine/Cosine term
 - Exponentially damped, time constant = $1/\alpha$
 - Oscillatory, period $T = \frac{2\pi}{\omega_d}$



Properties of Series RLC Network

- Behavior captured by damping
 - Gradual loss of the initial stored energy
 - α determines the rate of damping

- $\alpha > \omega_0$ (i.e., $R > 2\sqrt{\frac{L}{C}}$), overdamped

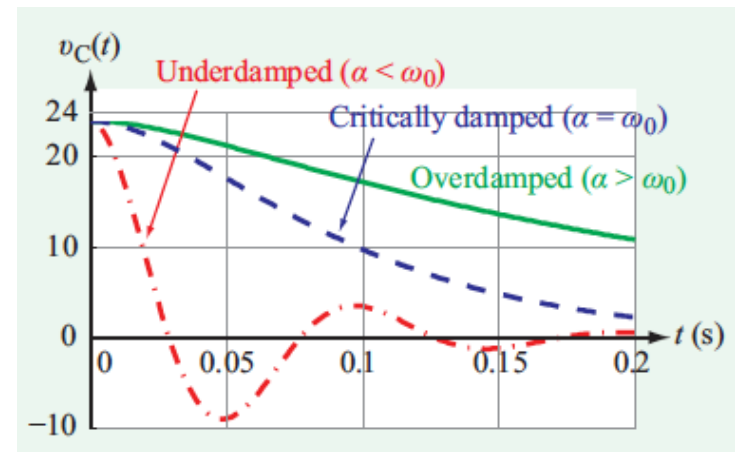
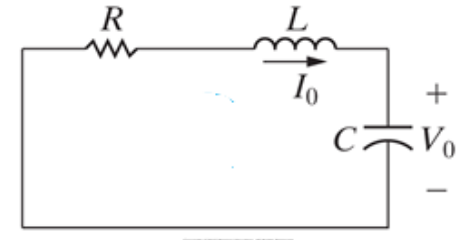
$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

- $\alpha = \omega_0$ (i.e., $R = 2\sqrt{\frac{L}{C}}$), critically damped

$$v(t) = (A_1 t + A_2) e^{-\alpha t}$$

- $\alpha < \omega_0$ (i.e., $R < 2\sqrt{\frac{L}{C}}$), underdamped

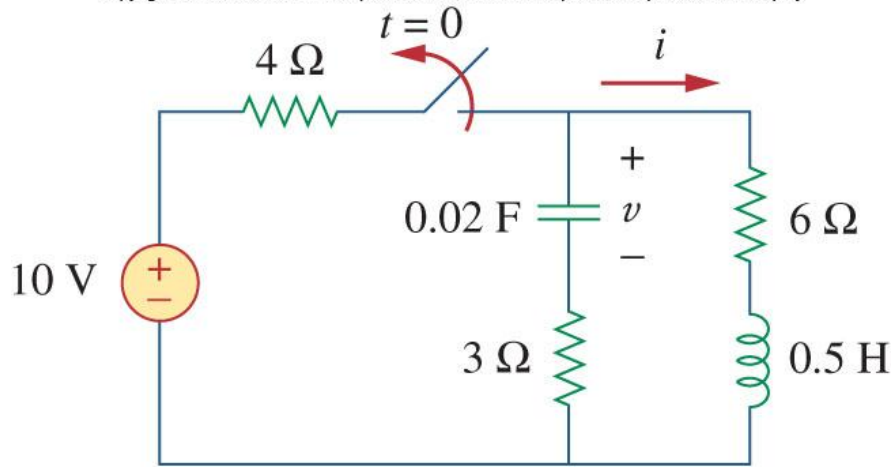
$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$



Example

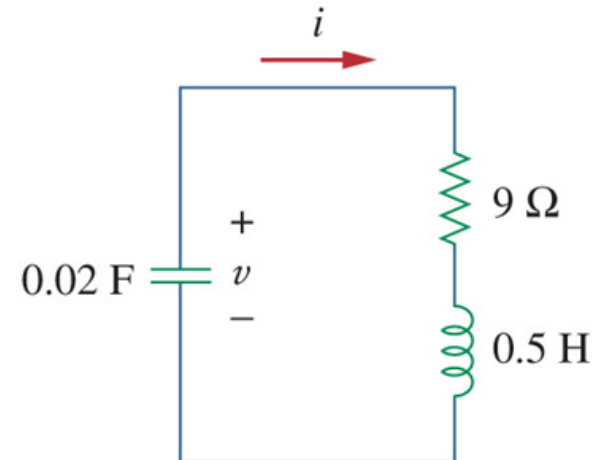
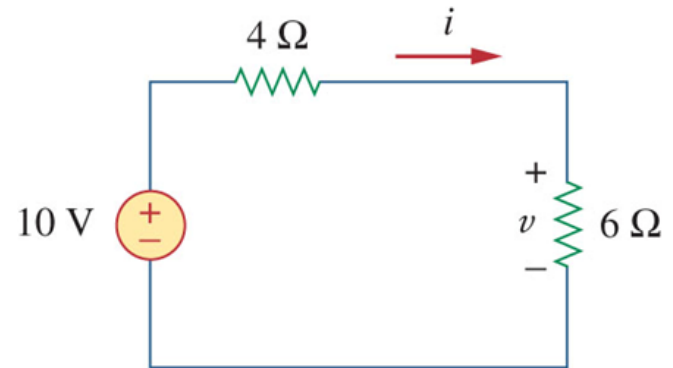
- Find $v(t)$ & $i(t)$ in the circuit below. Assume the circuit has reached steady state at $t = 0^-$.

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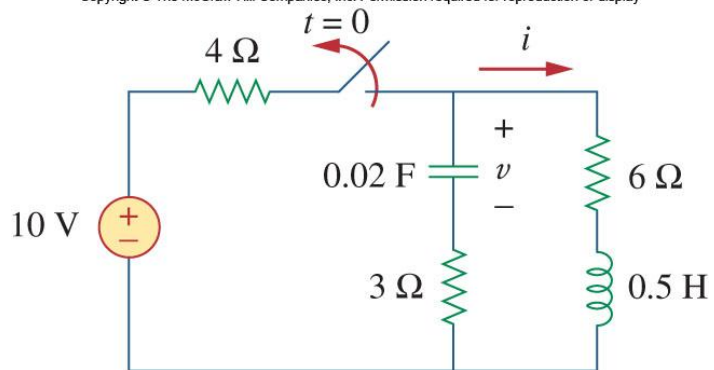
$$\alpha = \frac{R}{2L} = 9 \quad \omega_0 = \frac{1}{\sqrt{LC}} = 10$$

$$s_{1,2} = -\alpha \mp \sqrt{\alpha^2 - \omega_0^2} = -9 \mp j4.359$$





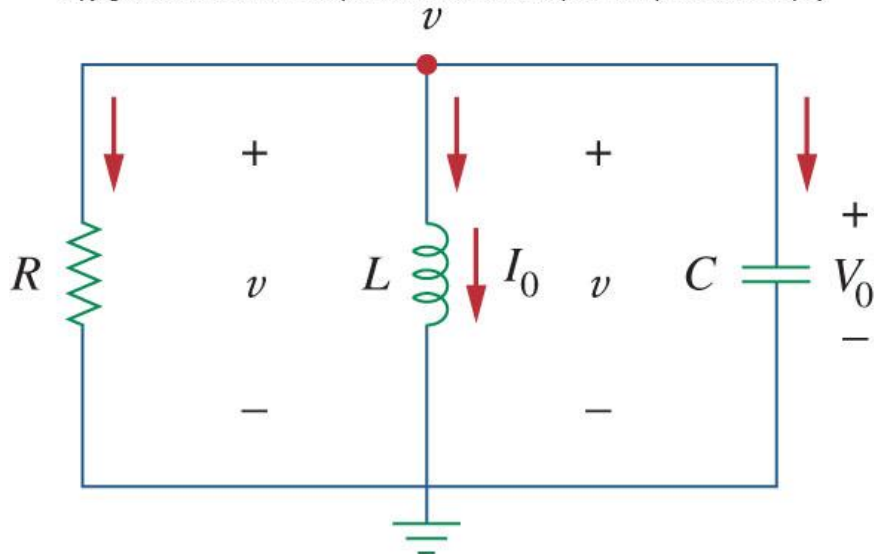
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Source-Free Parallel RLC Network

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Source-Free Parallel RLC Network

$$\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

- The characteristic equation is:

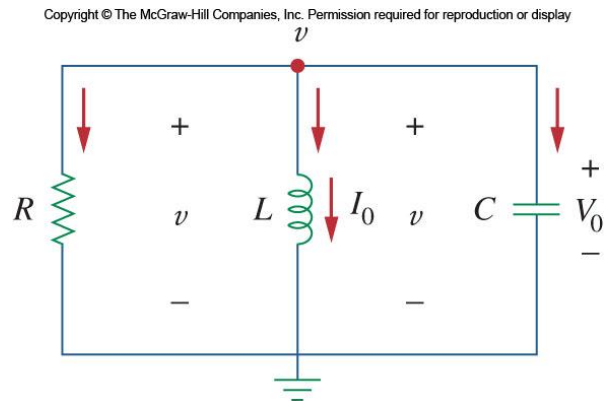
$$s^2 + \frac{1}{RC} s + \frac{1}{LC} = 0$$

From this, we can find the roots of the characteristic equation to be:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

As in last time, there are three scenarios to consider.



Three Damping Cases

$$\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

- For the overdamped case, the roots are real and negative,

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

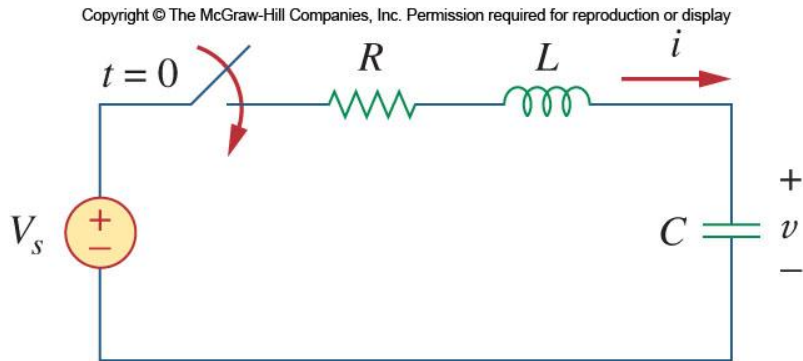
- For critically damped, the roots are real and equal

$$v(t) = (A_2 + A_1 t) e^{-\alpha t}$$

- In the underdamped case, the roots are complex

$$v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

Step Response of a Series RLC Circuit



$$\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$$

- The solution

$$v(t) = v_t(t) + v_{ss}(t)$$



The complete solutions for the three conditions of damping are:

- $v(t) = V_s + (A_1 e^{s_1 t} + A_2 e^{s_2 t})$ (Overdamped)
- $v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t}$ (Critically Damped)
- $v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$ (Underdamped)

Example

- Find $v(t)$ and $i(t)$ for $t > 0$.

Consider three cases:

- $R = 5\Omega$
- $R = 4\Omega$
- $R = 1\Omega$

When $R = 5\Omega$,

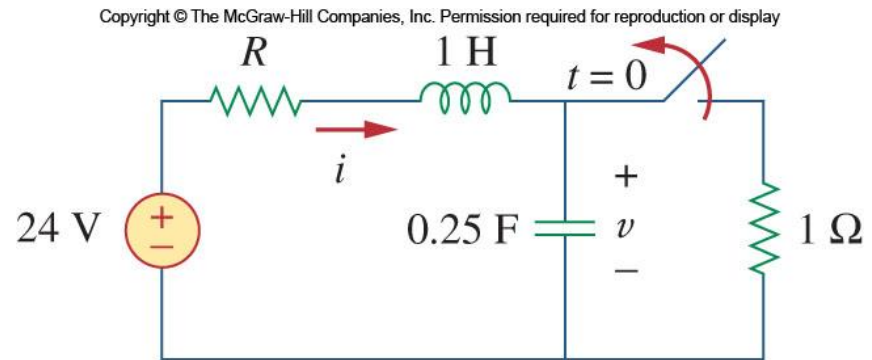
- For $t < 0$, switch closed, capacitor open, inductor shorted.

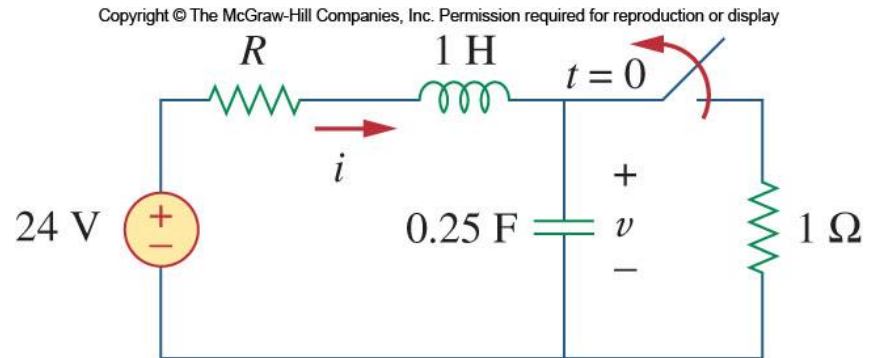
$$i(0) = 4A = C \frac{dv(0)}{dt}, \quad v(0) = 4V, \quad \frac{dv(0)}{dt} = 16$$

- For $t > 0$, switch open, a series RLC network

$$\alpha = \frac{R}{2L} = 2.5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 2, \quad s_{1,2} = -1, -4 \quad \text{Overdamped.}$$

$$v(t) = v_{ss} + (A_1 e^{-t} + A_2 e^{-4t})$$





When $R = 4\Omega$,

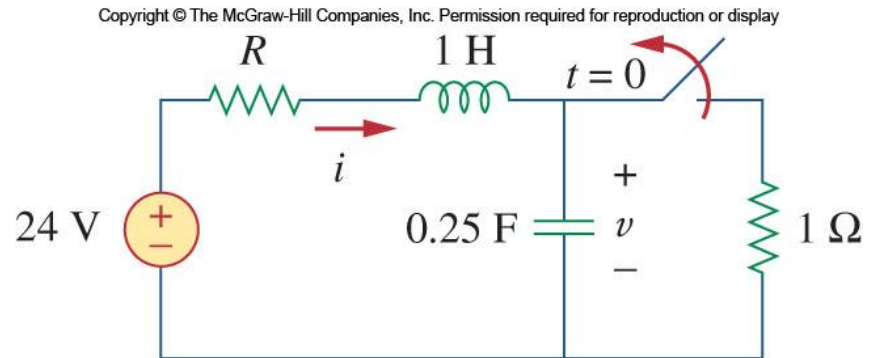
- For $t < 0$, switch closed, capacitor open, inductor shorted.

$$i(0) = 4.8A = C \frac{dv(0)}{dt}, \quad v(0) = 4.8V, \quad \frac{dv(0)}{dt} = 19.2$$

- For $t > 0$, switch open, a series RLC network

$$\alpha = \frac{R}{2L} = 2, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 2, \quad s_{1,2} = -2 \quad \text{Critically damped}$$

$$v(t) = v_{ss} + (A_1 + A_2 t)e^{-2t}$$



When $R = 1\Omega$,

- For $t < 0$, switch closed, capacitor open, inductor shorted.

$$i(0) = 12A = C \frac{dv(0)}{dt}, \quad v(0) = 12V, \quad \frac{dv(0)}{dt} = 48$$

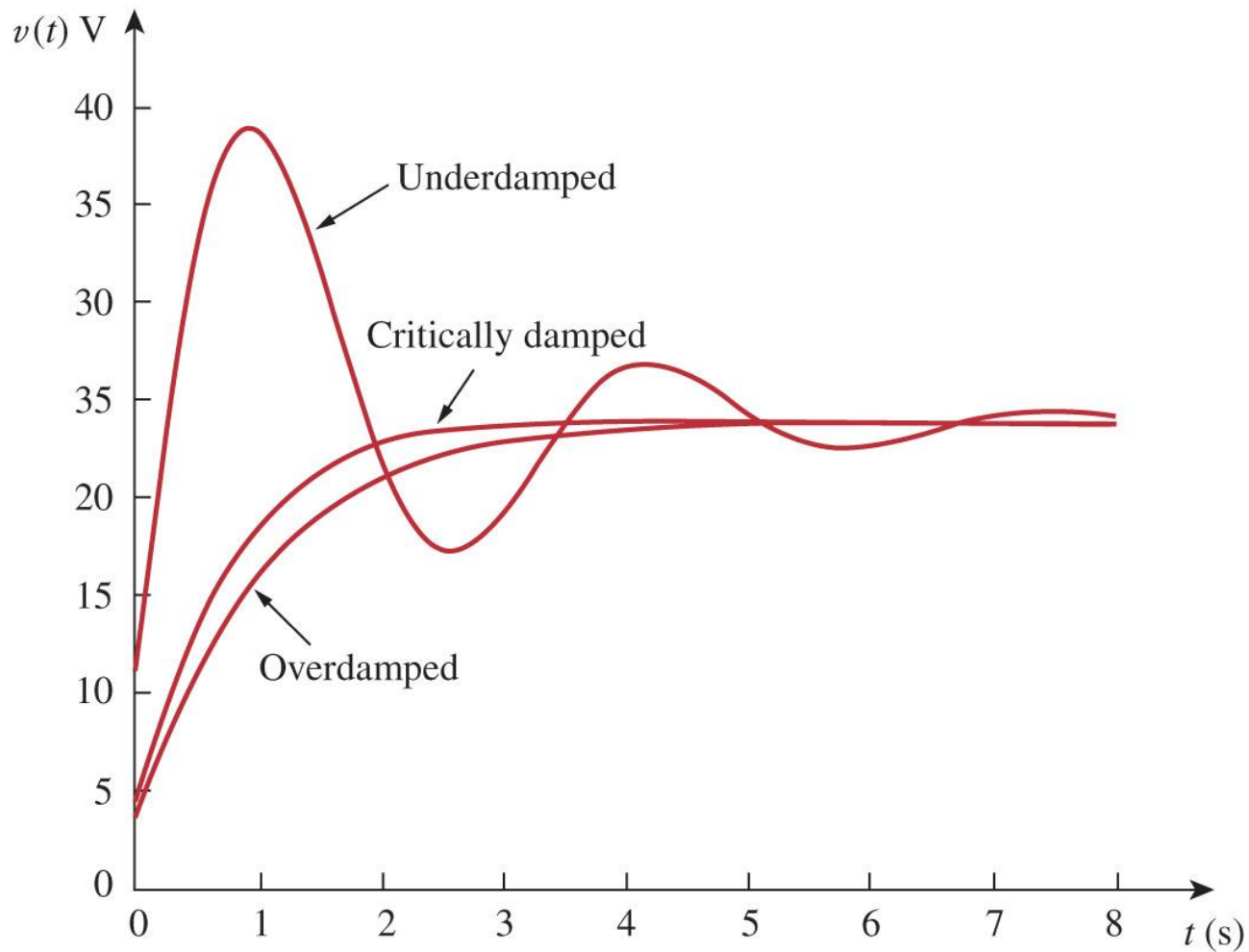
- For $t > 0$, switch open, a series RLC network

$$\alpha = \frac{R}{2L} = 0.5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 2, \quad s_{1,2} = -0.5 \mp j1.936 \quad \text{Underdamped}$$

$$v(t) = v_{ss} + (A_1 \cos 1.936t + A_2 \sin 1.936t)e^{-0.5t}$$

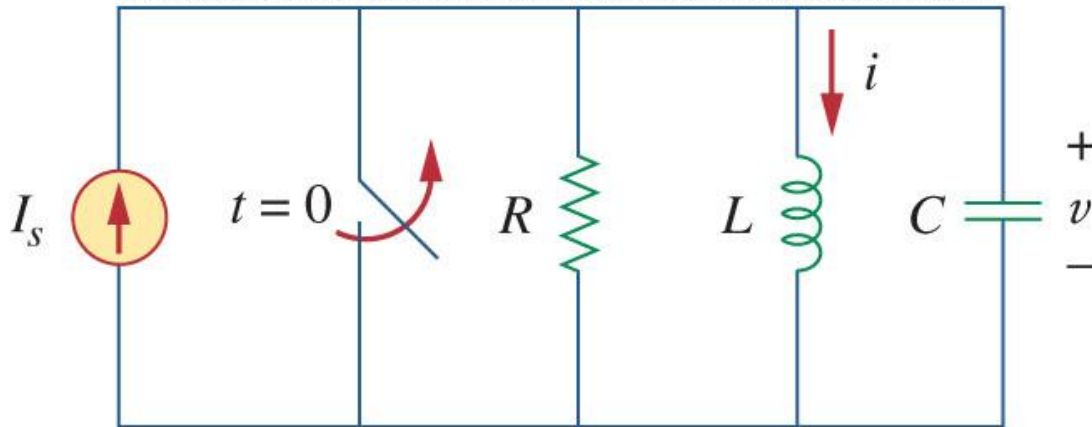


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Step Response of a Parallel RLC Circuit

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So we get

$$\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$$

Step Response of a Parallel RLC Circuit

$$\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$$

- As in the series RLC case, the response is a combination of transient and steady state responses:

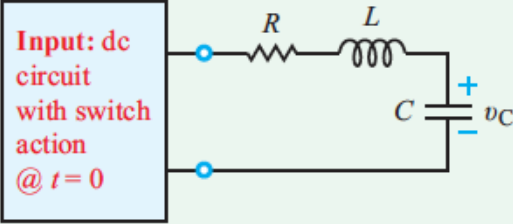
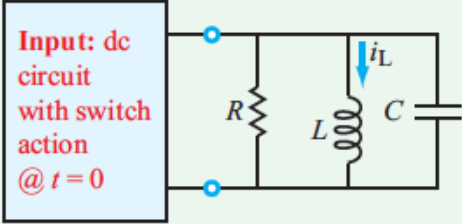
$$i(t) = I_s + A_1 e^{\tau_1 t} + A_2 e^{\tau_2 t} \quad (\text{Overdamped})$$

$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically Damped})$$

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

Here the variables A_1 and A_2 are obtained from the initial conditions, $i(0)$ and $di(0)/dt$.



Series RLC 	Parallel RLC 
Total Response	Total Response
Overdamped ($\alpha > \omega_0$) $v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$ $A_1 = \frac{\frac{1}{C} i_C(0) - s_2 [v_C(0) - v_C(\infty)]}{s_1 - s_2}$ $A_2 = \left[\frac{\frac{1}{C} i_C(0) - s_1 [v_C(0) - v_C(\infty)]}{s_2 - s_1} \right]$	Overdamped ($\alpha > \omega_0$) $i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$ $A_1 = \frac{\frac{1}{L} v_L(0) - s_2 [i_L(0) - i_L(\infty)]}{s_1 - s_2}$ $A_2 = \left[\frac{\frac{1}{L} v_L(0) - s_1 [i_L(0) - i_L(\infty)]}{s_2 - s_1} \right]$
Critically Damped ($\alpha = \omega_0$) $v_C(t) = (B_1 + B_2 t) e^{-\alpha t} + v_C(\infty)$ $B_1 = v_C(0) - v_C(\infty)$ $B_2 = \frac{1}{C} i_C(0) + \alpha [v_C(0) - v_C(\infty)]$	Critically Damped ($\alpha = \omega_0$) $i_L(t) = (B_1 + B_2 t) e^{-\alpha t} + i_L(\infty)$ $B_1 = i_L(0) - i_L(\infty)$ $B_2 = \frac{1}{L} v_L(0) + \alpha [i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$) $v_C(t) = e^{-\alpha t} (D_1 \cos \omega_d t + D_2 \sin \omega_d t) + v_C(\infty)$ $D_1 = v_C(0) - v_C(\infty)$ $D_2 = \frac{\frac{1}{C} i_C(0) + \alpha [v_C(0) - v_C(\infty)]}{\omega_d}$	Underdamped ($\alpha < \omega_0$) $i_L(t) = e^{-\alpha t} (D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$ $D_1 = i_L(0) - i_L(\infty)$ $D_2 = \frac{\frac{1}{L} v_L(0) + \alpha [i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations $\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$ $s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$ $\omega_0 = \frac{1}{\sqrt{LC}}$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ $s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$	



***8.21** Calculate $v(t)$ for $t > 0$ in the circuit of Fig. 8.75.

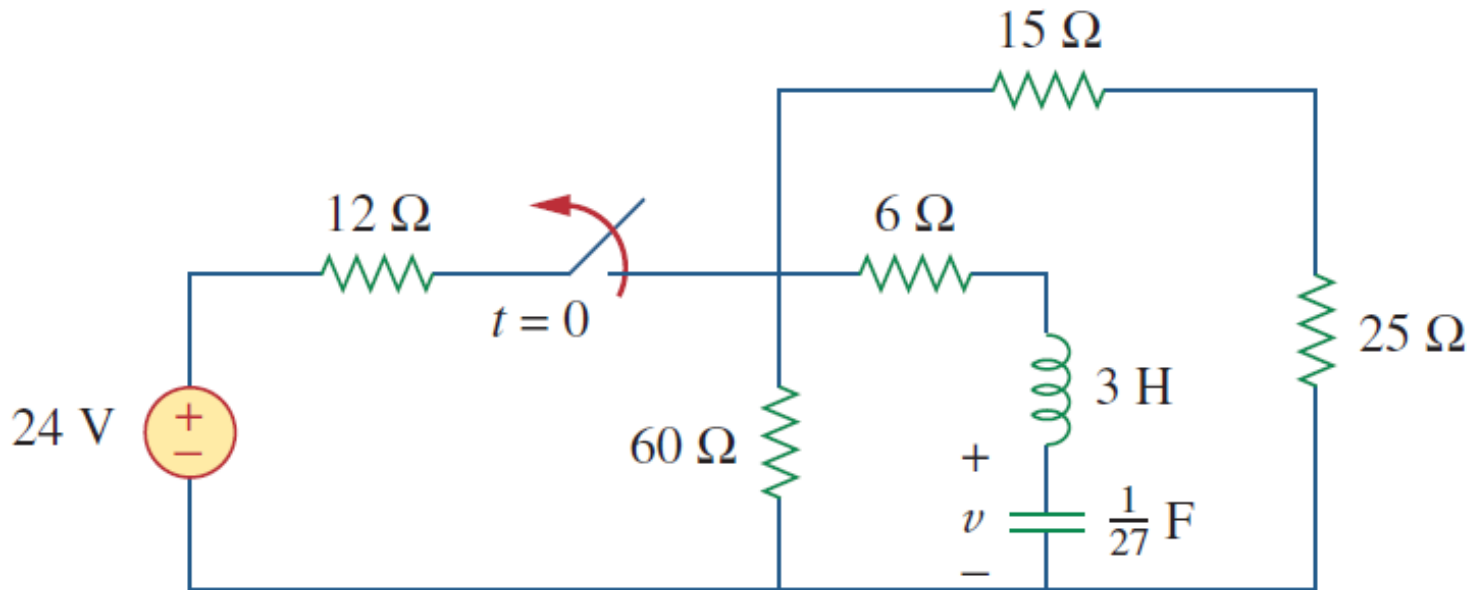


Figure 8.75
For Prob. 8.21.

8.49 Determine $i(t)$ for $t > 0$ in the circuit of Fig. 8.96.

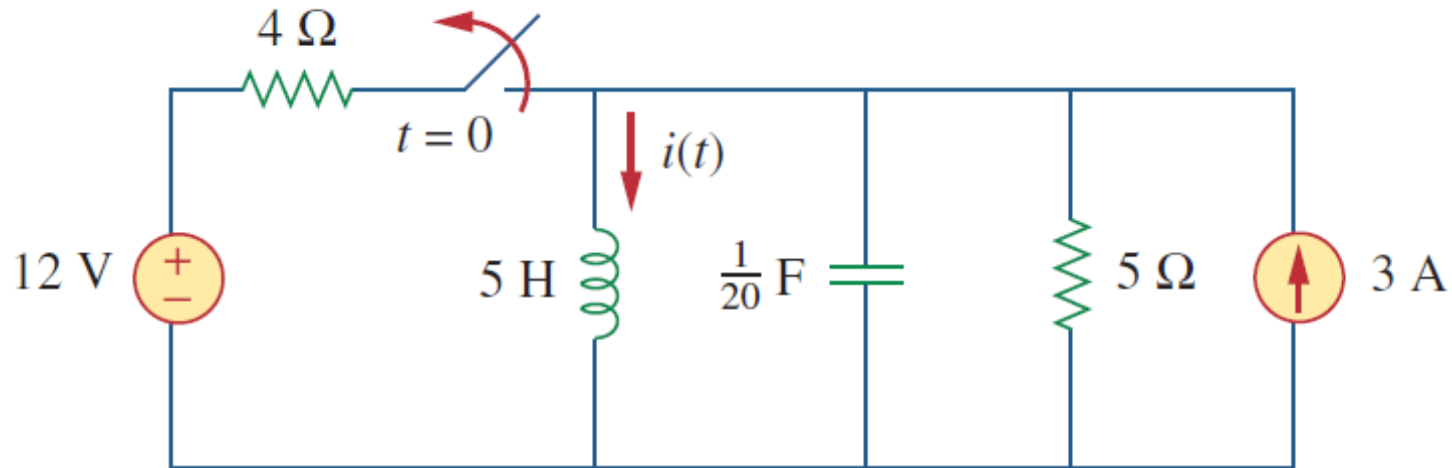


Figure 8.96

For Prob. 8.49.



General Second-Order Circuits

- The principles of the approach to solving the series and parallel forms of RLC circuits can be applied to general second order circuits:
 1. First determine the initial conditions, $x(0)$ and $dx(0)/dt$.
 2. Find the equation
 3. The total response = transient response + steady-state response.

$$x(t) = x_t(t) + x_{ss}(t)$$

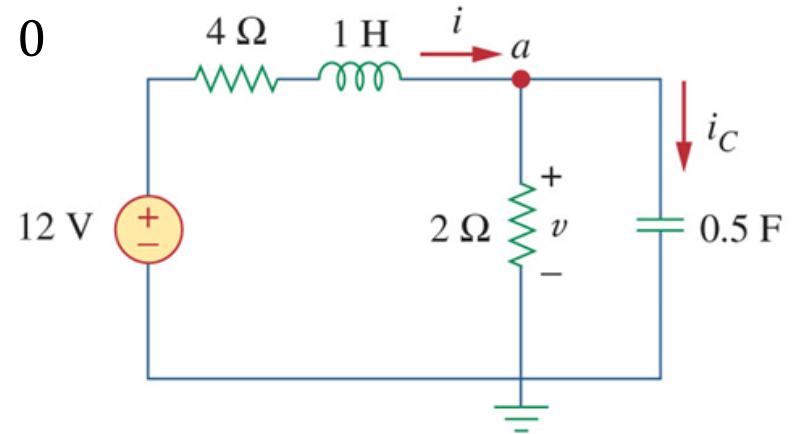
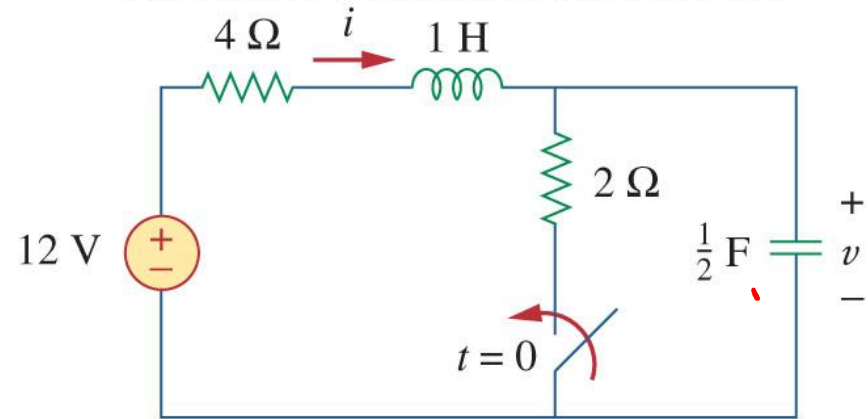


General RLC Circuits

- Find the complete response v for $t > 0$ in the circuit.

1. Initial conditions

$$v(0^+) = v(0^-) = 12V, i(0^+) = i(0^-) = 0$$



Example of 2nd-order op-amp circuits $C_2 = 100\mu F$

- Find v_o for $t > 0$ when $v_s = 10u(t)mV$.

KCL at node 1:

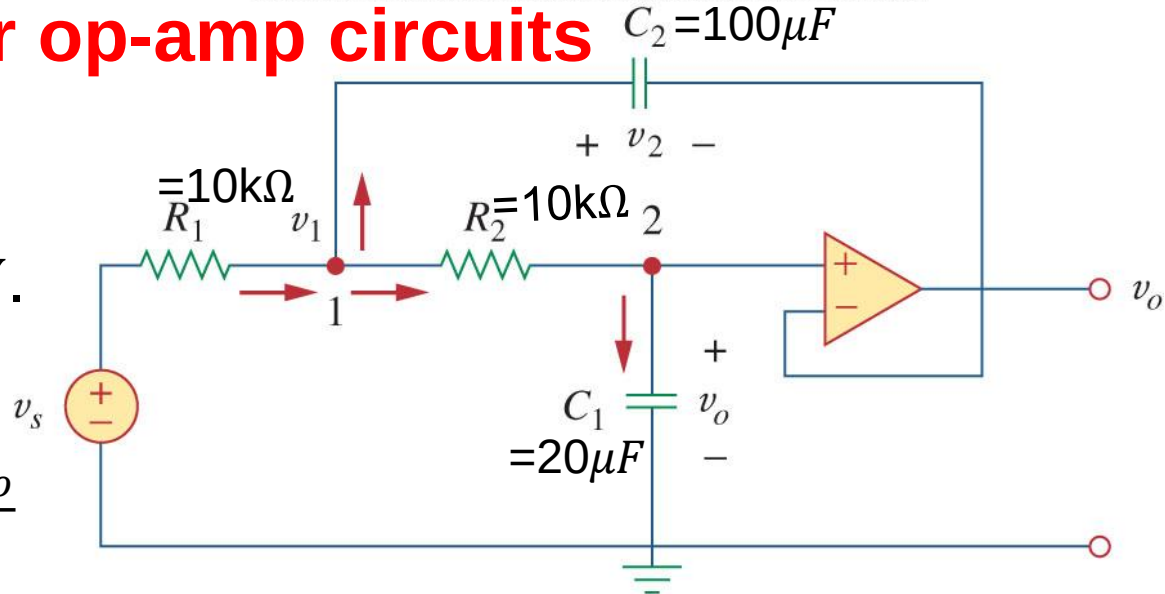
$$\frac{v_s - v_1}{R_1} = C_2 \frac{dv_2}{dt} + \frac{v_1 - v_o}{R_2}$$

KCL at node 2:

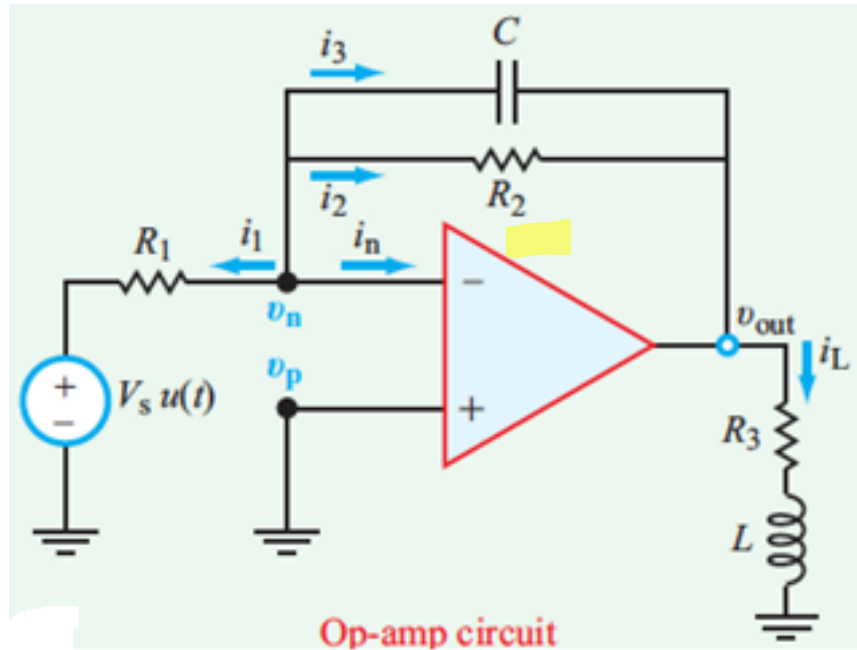
$$C_1 \frac{dv_o}{dt} = \frac{v_1 - v_o}{R_2} \quad v_1 - v_2 = v_o$$

$$\Rightarrow \frac{d^2 v_o}{dt^2} + \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} \right) \frac{dv_o}{dt} + \frac{v_o}{R_1 R_2 C_1 C_2} = \frac{v_s}{R_1 R_2 C_1 C_2}$$

Initial conditions: $v_o(0^+) = 0, C_1 \frac{dv_o(0^+)}{dt} = \frac{v_1(0^+) - v_o(0^+)}{R_2} = \frac{v_2(0^+)}{R_2} = 0$



Example



$$\frac{R_3}{R_2} i_L + \left(\frac{L}{R_2} + R_3 C \right) \frac{di_L}{dt} + LC \frac{d^2 i_L}{dt^2} = -\frac{V_s}{R_1}$$

$$i_L(\infty) = \frac{v_{out}(\infty)}{R_3} = -\frac{R_2 V_s}{R_1 R_3} = -1 \text{ mA}$$

$$i_L(0) = i_L(0^-) = 0, \quad i_L'(0) = \frac{1}{L} v_L(0) = 0.$$



2nd-order Oscillation Circuit

